Lee Wei Kiat

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EDUCATION

National University of Singapore (NUS) Aug 2023 - Dec 2026
Bachelor of Science in Business Analytics
- GPA: 4.78/5.00 - Honours (Highest Distinction), Specialization in Machine Learning-based Analytics

Ngee Ann Polytechnic Apr 2018 - Apr 2021
Diploma in Financial Informatics (Merit)
- GPA: 3.93/4.00, Directors List (2018-2020), Specialization in Financial Analytics

WORK EXPERIENCE

ALPS Pte Ltd, Data Analytics and Information Systems Intern Jun 2025 - Dec 2025
- Facilitated data onboarding onto the HEALIX analytics platform (Databricks), conducting schema validation and end-to-end data parity checks against legacy sources.
- Conducted data parity checks across stakeholder Tableau dashboards, verifying metric consistency and zero visual or calculation drift.
- Maintained stakeholder Tableau dashboards, resolving ad-hoc enhancement requests and updating calculations, filters, and visuals to align with changing reporting needs.

Deloitte, Risk Advisory Intern Sep 2020 - Feb 2021
- Performed journey testing on the client's data using ACL to assess process accuracy.
- Identified and analyzed potential risk areas to recommend mitigation strategies.
- Evaluated the effectiveness of the client's system controls, contributing to compliance and security.

PROJECTS

NCCS Capstone Project (ANCHOR Chatbot), AI Engineer Jan 2026 - Apr 2026
Local-first agentic reasoning system for natural-language clinical queries on OMOP CDM data
- Architected an 8-agent LangGraph workflow executing locally via Ollama within an 8 GB GPU VRAM footprint, orchestrating DeepSeek-R1 (reasoning), and Qwen2.5-Coder (SQL/stats)
- Implemented dynamic model VRAM management (keep_alive lifecycle) and a Data Registry LRU cache using serialized data reference keys to prevent inter-node memory exhaustion during analytical execution.
- Built a hybrid concept-resolution engine using Bio_ClinicalBERT and FAISS, with dataset-level fallbacks to eliminate OMOP Concept ID hallucinations in generated SQL.
- Designed an autonomous DuckDB SQL generation pipeline featuring dynamic YAML schema pruning, column-resolution gating, and a 4-retry self-correction loop fed with live query execution traces.
- Engineered a deterministic statistical layer executing verified SciPy and lifelines routines (Kaplan-Meier, Cox regression, chi-squared) with a human-in-the-loop clarification gate for ambiguous clinical definitions.

Judicial Analytics, Data Scientist Jan 2026 - Apr 2026
Predicting defendant liability in Singapore corporate litigation using deep learning.
- Engineered a 3-stage GPT pipeline with an LLM-as-a-Judge auditor, converting 122 Singapore judgments into leakage-safe, ML-ready JSON via a pre-dispute "temporal wall" and regex scrubbing.
- Built a Hierarchical Attention Network (HAN) with SwiGLU over chronologically ordered facts; benchmarked vs. XGBoost and Legal-BERT using a temporal split and case-level F1/Recall.

P2P Loan Default Prediction System, Data Scientist Jan 2025 - Apr 2025
Built different models prioritizing a high recall rate to detect potential defaulters.
- Processed 2.9 mil loan records using PySpark and leveraged PCA to reduce features (83 to 62).
- Addressed class imbalances using SMOTE and ENN; trained LR, RF, SVM, and GBT classifiers.

TECHNICAL SKILLS

- **Generative AI & Agents:** LangGraph, Ollama, FAISS, Hugging Face (Transformers), Prompt Engineering, Agentic Workflows.
- **Machine Learning & Data Science:** Deep Learning (BERT, HAN, LSTMs), Scikit-Learn, PySpark, Statistical Modeling (SciPy, lifelines), Feature Engineering.
- **Programming & Frameworks:** Python, SQL, Java, C#, JavaScript, Vue.js, HTML/CSS.
- **Data Engineering & BI:** Databricks, Tableau, Data Validation & QA, Schema Mapping, ETL / Data Pipelines.
- **Developer Tools & Platforms:** Git, Docker, AWS, Azure, MLFlow.